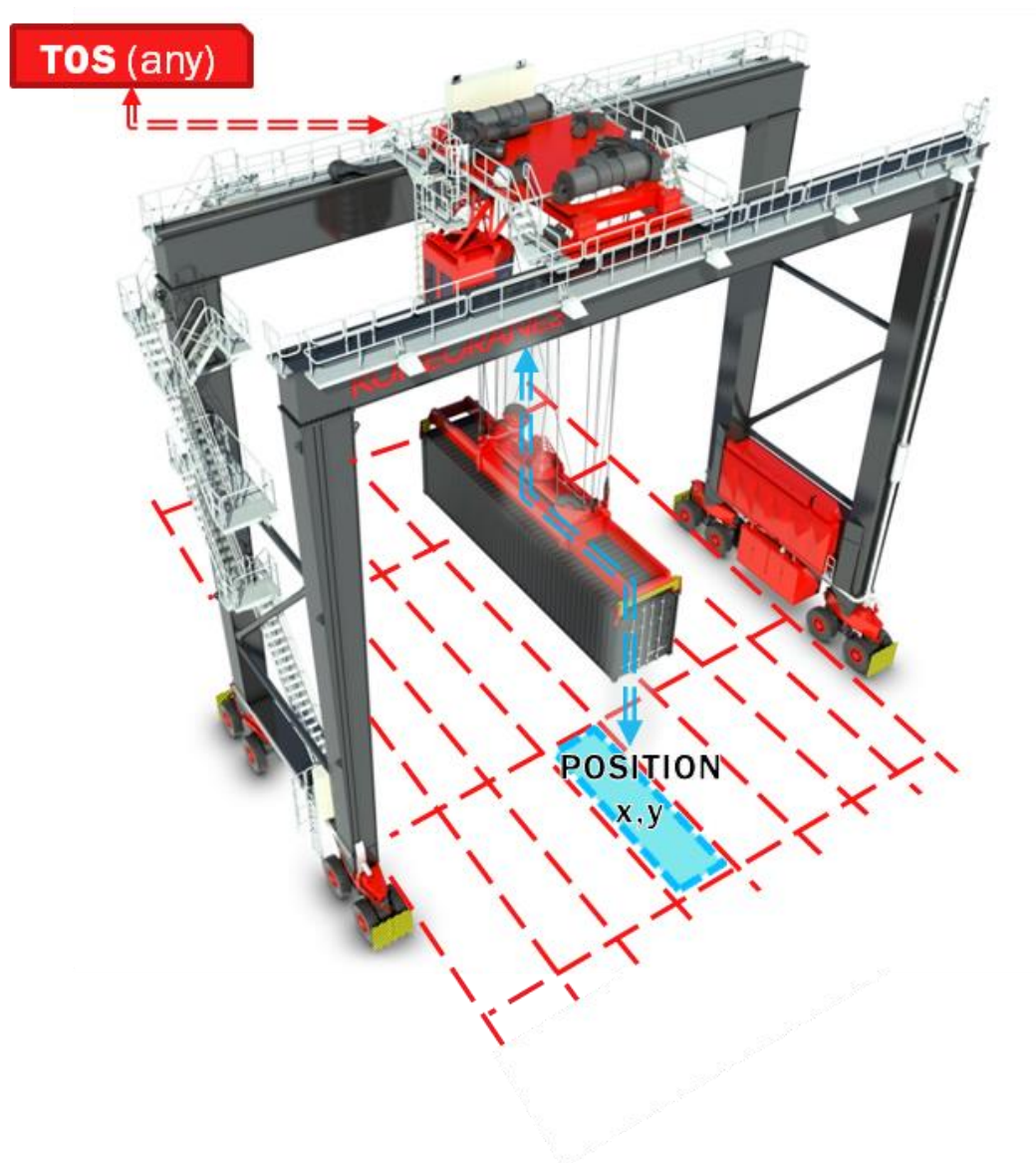


INTERFACE TO THE TOS

Technical Description

Yard Process Automation
21.04.2015
3.3



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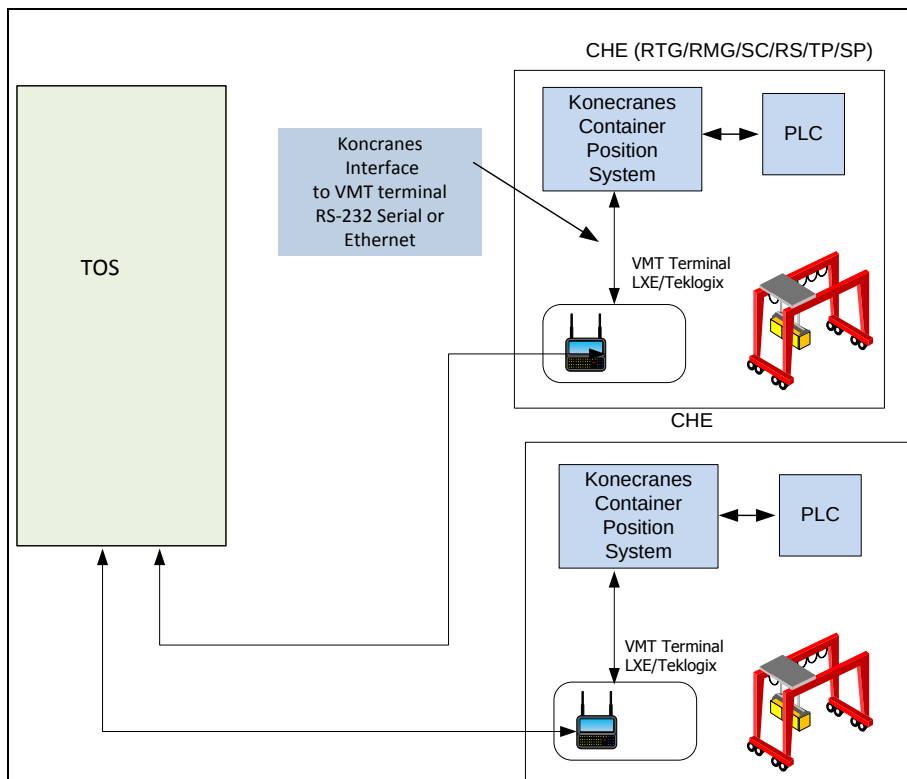
1. TERMINOLOGY

Auto-TOS	Konecranes Auto-TOS reporting system
Auto-pos	Konecranes Auto-positioning system
CHE	Container Handling Equipment
CRC	Cyclic Redundancy Counter
TOS	Terminal Operating System
VMT	Vehicle Mount Terminal

2. GENERAL DESCRIPTION

The interface to the TOS provides real-time messaging between the Konecranes Auto-TOS reporting system and the TOS. The messages are a mechanism for updating the TOS with real-time container movement information. The messaging is bi-directional: from the Auto-TOS to the TOS, and from the TOS to the Auto-TOS.

The interface to the TOS is implemented using an RS-232 serial connection port or Ethernet port in the Auto-TOS. The connection to the TOS terminal is a direct connection. Alternatively, a fiber optical cable can be used between the Konecranes Auto-TOS and TOS terminals, depending on the distance between the terminals. When Ethernet communication is used, the VMT terminal is connected to crane network



3. ESTABLISHING THE CONNECTION

A connection can be established without a protocol. The Auto-TOS sends pick and place messages automatically after power-on. The Auto-TOS sends a heartbeat message once per second (the time period is configurable), so that the TOS can detect any communication failures. The heartbeat message can also be requested from the TOS, and automatic heartbeat sending can be disabled.

4. MESSAGE ENCAPSULATION

The following protocol is used to encapsulate messages within the byte-stream in order to e.g. detect the beginning and end of a message. The protocol uses two length bytes and a termination byte. Thus the format of the transmitted packet is:

FFXXNNaaaaaaaaaaaa...aaZZ

Where:

- "FF" indicates "start of data" two byte value hex 0xFFFF (internal use)
- "XX" denotes message length by two byte asciihex, max 255 bytes
- "NN" is a message ID code (Msg-ID)
- "NNaaaaaaaaaaaa...aa" is the message as defined later
- "ZZZZ" is the CRC checksum. All of the bytes in the message, including the "XX", will be added up into a binary word that is two bytes wide, two's complemented. ($\sim\text{sum}+1$). The result is transferred to Asciihex format.

Note 1: The message length "XX" is defined as the length of the "NNaaaaaaaaaaaa...aa" part.

Note 2: Ascii fields are left-justified and padded with blanks unless otherwise noted. Asciihex fields used for 8-bit and 16-bit Integers are in big-endian format (Message Length, Counter, CRC, etc.).

5. AUTO-TOS TO TOS MESSAGES

A1	Heartbeat	0x4131
A2	Pick/Place Message	0x4132
A3	TOS Message Acknowledgement	0x4133
A4	Job Twistlock Action Request	0x4134

6. TOS TO AUTO-TOS MESSAGES

B1	Heartbeat Request	0x4231
B2	Auto-TOS Message Acknowledgement	0x4232
B3	Job Data Message	0x4233
B4	Job Cancel Message	0x4234
B5	Job Continue Message	0x4235

7. MESSAGES FROM AUTO-TOS TO TOS FORMAT DESCRIPTIONS

7.1. Heartbeat Message (A1) format description

Offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("A1"=0x4131 = 16689)
002	6	Ascii	Version	Version number
008	6	Ascii	Time	Read time (HHMMSS) E.g.: 1:31 PM is represented as 133100
014	4	Ascii	Status	Auto-steering status (OK = "0000", any error: <> 0 (d))
018	8	Ascii	Position /	Block name /"ERROR" /"NO BLOCK" bytes 018 – 025
026	3			Bay number bytes 026 – 028
029	3			Row number bytes 029 – 031
032	4			Height, Logical value 1...7 bytes 032 – 035
036	2	Ascii	Crane status	"01" = Crane off "02" = Crane on /idle "03" = Crane running "04" = Crane error / Fatal error
038	2	Ascii	GPS status	"01" = GPS O.K "02" = GPS internal module fault "03" = No satellites "04" = No communication to base station
040	5	Ascii	CHE	CHE id
045	6	Ascii	System status	Reserved for internal use System status information
051	2	Ascii	PLC comm. status	"01" = Communication OK "02" = Communication Failure
053	9	Ascii	PosX	Che position x-coordinate
062	9	Ascii	PosY	Che position y-coordinate
071	2	Ascii	Len	Spreader size 20/30/40/45/T (T=Twin-lift)
073	1	Ascii	Twist lock	"O" twist lock open "C" twist lock close

Note 3: Active GPS status values depends on the CHE type and hardware configuration

Note 4: Status and Crane Status used only with auto-steering RTG.

Note 5: Status, Cranes status, GPS status, System Status and PLC comm. Status are used mainly for Konecranes diagnostic purposes.

Note 6: Height is specified as logical height values '1' to '7'. Konecranes YardIT ContPos reads the height value from PLC and converts it to a logical value. Left justified space filled.

Note 7: Position information:

The positions are represented in meters from base station (origo).

X-coordinate increases down on the map and Y-coordinate increases right on the map.

The map is rotated "straight" before calculating these coordinates.

18 bytes reserved for X/Y-position.

9 bytes reserved for both X and Y position.

9 bytes are:

(-)DDDDD.DD (two decimal precision i.e one centimeter which is enough)

The positions can vary from -99999.99 m - 99999.99 meters

examples

-1.24

23.45

127.45

-1256.23

-15674.45

etc.

The strings are padded with spaces in front of the string

7.2. Pick/Place Message (A2) format description

offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("A2" = 0x4132=16690)
002	6	Ascii	Version	Version number
008	2	AsciiHex	Counter	0-255
010	2	Ascii	Move type	03" = CHE lifts container from ground 04" = CHE grounds container
012	14	Ascii	Time	Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM on 3 of July 2004 is represented as 20040703133100
026	5	Ascii	Status	Auto-steering status (OK = "0000", any error: <> 0 (d))
031	5	Ascii	CHE	CHE id
036	5	Ascii	ContWeight	Container weight in kilograms. Valid in Place mes- sage only
041	2	Ascii	PosOnTruck	Optional. Measured position when picking/placing from truck. Measured in spreader up position
043	5	Ascii	spare	[spaces]
048	12	Ascii	ContID1	Optional
060	12	Ascii	ContID2	Optional
072	1	Ascii	Mount	Always "Y"
073	1	Ascii	Tag	Always "N"
074	8	Ascii	Position1	Block name /"ERROR " /"NO BLOCK"
082	3			Bay number
085	3			Row number
088	4			Height
092	8	Ascii	Position2	Block name /"ERROR " /"NO BLOCK"
100	3		(Filled only for	Bay number
103	3		Twin-lift)	Row number
106	4			Height
110	2	Ascii	Len	Container size 20/30/40/45/T (T=Twin-lift)
112	6	Ascii	System status	Reserved for internal use System status information

This message is sent automatically by the Auto-TOS whenever the Container Handling Equipment picks a container up or puts one down. The message format has some optional/extra fields for future expansion, unused at the moment...

For twin-lift operation, the container length field (Len) will be "T", and the Position1 and Position2 fields are both filled. If only one container is handled, the container length field (Len)

will be 20 or 40 (feet) and only the Position1-field is filled. The Konecranes Auto-TOS does not use container sizes 30 and 45 at the moment.

Konecranes ContPos resends the Pick/Place message if it does not receive B2 acknowledgement message with time limit. Resending can be disabled in the ContPos configuration.

7.3. TOS Message Acknowledgement (A3) format description

offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("A3"=0x4133=16691)
002	2	AsciiHex	Counter	0-255

Every TOS message B3, B4 and B5 will receive an A3-acknowledgement message. The counter equals the received message counter and identifies the message to be acknowledged.

If the TOS does not receive an acknowledgement within 5 seconds, it should resend the message again.

After 3 retries without a response, a "NO communication with Konecranes system" alarm shall be reported by the TOS.

Counter: The TOS will incrementally raise the counter value before every new message, but not for retries. The Konecranes Auto-TOS reporting system will check if the message's "Counter" field was incrementally changed and will deal with it as a new message only when it has changed.

7.4. Job Twist lock Action request (A4) format description

offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID (" A4 " = 0x4134=16692)
002	6	Ascii	Version	Version number
008	2	AsciiHex	Counter	0-255
010	2	Ascii	Move type	"03" = CHE lifts request from ground "04" = CHE container ground request
012	14	Ascii	Time	Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM on 3 of July 2004 is represented as 20040703133100
026	5	Ascii	Status	Auto-steering status (OK = "0000", any error: <> 0 (d))
031	5	Ascii	CHE	CHE id
036	12	Ascii	Chassis	[spaces]
048	12	Ascii	ContID1	Optional
060	12	Ascii	ContID2	Optional
072	1	Ascii	Mount	Always "Y"
073	1	Ascii	Tag	Always "N"
074	8	Ascii	Position1	Block name /"ERROR " /"NO BLOCK"
082	3			Bay number
085	3			Row number
088	4			Height
092	8	Ascii	Position2	Block name /"ERROR " /"NO BLOCK"
100	3		(Filled only for	Bay number
103	3		Twin-lift)	Row number
106	4			Height

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110	2	Ascii	Len	Container size 20/30/40/45/T (T=Twin-lift)
112	6	Ascii	System status	Reserved for internal use System status information

This message is sent automatically by the Auto-TOS every time the Container Handling Equipment tries to pick or place a container with active twist-lock blocking. The message structure is the same as in the pick/place message. Normally the response will be a Job Continue message which unblocks the twistlocks.

8.MESSAGES FROM TOS TO AUTO-TOS FORMAT DESCRIPTIONS

8.1. Heartbeat request (B1) format description

offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("B1" = 0x4231=16945)
002	1	Ascii	TWL_block	"0" = Twist locks unblocked "1" = Twist locks unblocked in truck lane "2" = Twist locks blocking active

The TOS application can control the twistlock blocking/unblocking status when sending the Heartbeat Request. The Auto-TOS sends an A1 message as a response. Normally, the Auto-TOS automatic heartbeat sending function is disabled in this case. Twistlock blocking control can be bypassed momentarily by sending a Job Continue message.

8.2. Auto-TOS Message Acknowledgement (B2) format description

offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("B2" = 0x4232=16946)
002	2	AsciiHex	Counter	0-255

Every A2 and A4 message will receive a B2 acknowledgement message from the TOS. If the Auto-TOS does not receive an acknowledgement within 2 seconds (time is configurable), it will resend the message.

After 3 retries without a response, a "NO Communication with TOS system" alarm will be reported to the crane's PLC (RTG).

A new A2 message will be sent in response to every new message from the TOS.

When an acknowledgement message is received – the "NO Communication with TOS system" alarm will be turned off and a report will be sent to the crane's PLC (RTG).

Counter: The Konecranes Auto-TOS will incrementally raise the counter value before every new message, except with retries. The TOS will check if the message's "Counter" field was incrementally changed and will deal with it as a new message only if it has changed. If the counter stays the same, it means that the Auto-TOS did not receive the acknowledgement message: no new e.g. pick/place action was detected.

8.3. Job Data Message (B3) format description

Offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("B3" = 0x4233=16947)
002	6	Ascii	Version	Version number
008	2	AsciiHex	Counter	0-255
010	14	Ascii	Time	Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM on 3 of July 2004 is represented as 20040703133100
024	5	Ascii	CHE	CHE id
029	12	Ascii	ContID1	Container ID
041	12	Ascii	ContID2	Container ID 2 (Filled only for Twin-lift)
053	8	Ascii	Lift-Position1	Block name bytes 053 – 060
061	3			Bay number bytes 061 – 063
064	3			Row number bytes 064 – 066
067	4			Height, Logical value 1...7 bytes 067 – 070
071	8	Ascii	Lift-Position2 (Filled only for Twin-lift)	Block name bytes 071 – 078
079	3			Bay number bytes 079 – 081
082	3			Row number bytes 082 – 084
085	4			Height, Logical value 1...7 bytes 085 – 088
089	8	Ascii	Place-Position1	Block name bytes 089 – 096
097	3			Bay number bytes 097 – 099
100	3			Row number bytes 100 – 102
103	4			Height, Logical value 1...7 bytes 103 – 106
107	8	Ascii	Place-Position2 (Filled only for Twin-lift)	Block name bytes 107 – 114
115	3			Bay number bytes 115 – 117
118	3			Row number bytes 118 – 120
121	4			Height, Logical value 1...7 bytes 121 – 124
125	2	Ascii	Len	Container size 20/30/40/45/T (T=Twin-lift)
127	2	Ascii	PosOnTruck	Container position on truck or planned position to truck starting from back of the truck 1. 20" rear position 2. 20" middle position or (40/45 container) 3. 20 " front position

This message is sent by the TOS every time the Container Handling Equipment (CHE) driver selects a job, and the job is assigned to the CHE. This data is needed for automated CHE operations (e.g. Auto-Stop).

The Lift-Position1, Lift-Position2, Place-Position1 and Place-Position2 fields have the same format as Position1 and Position2 in the Pick/Place message. **The container length field (Len) is important for Auto-positioning: it's needed to calculate the stopping position correctly (also for twin-lift operation).** For twin-lift operation, the container length field (Len) is "T" and all four Position fields are filled. If only one container is handled, only the Lift-Position1 and Place-Position1 fields are used.

8.4. Job Cancel Message (B4) format description

Offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID ("B4" = 0x4234=16948)
002	2	AsciiHex	Counter	0-255

The TOS can cancel an ongoing job by sending the Job Cancel Message.

Acknowledgement is required. Job can't be cancelled if the twistlocks are engaged and container has been picked. A PICK message has been send to TOS. .

8.5. Job Continue Message (B5) format description

Offset	#bytes	Type	Name	Description
000	2	Ascii	Msg-ID	Two byte msg ID (" B5 " = 0x4235=16949)
002	2	AsciiHex	Counter	0-255

The Job Continue Message communicates that the current job state can continue.
Acknowledgement is required.

9.INDICATION BASED ON HEARTBEAT GPS STATUS

GPS status

"0001" = GPS OK (Green indication, system OK)

"0002" = GPS internal module fault (red indication, Alarm)

"0003" = No satellites (red indication, Alarm)

"0004" = No communication to base station (red indication, Alarm)

10. VALUES OF "POSITION" FIELDS

The Auto-TOS must conform to the customer's names for the position. The yard mapping and Auto-TOS configuration specify the valid values for each position field.

10.1. Position notation

Y12 , 250, 001, 1

- Where the Y12 **Area/Block** code is left-justified and padded with 5 spaces
- The Bay number max 3 characters, left-justified and space-padded
- The Row number max 3 characters left-justified and space-padded
- The height (Tier) is defined as logical value 1 to 7 left-justified and space-padded

10.2. "No block" in position field

"NO BLOCK" is announced if the Container Handling Equipment is out of a defined area: Out of block, bay and row.

10.3. "Error" in the position field

If "GPS OK" is not displayed as the GPS status, or if there is a height measurement error, the text "ERROR " (8 characters) is displayed in the "Block name" bytes. The remaining position bytes will be filled with spaces. This means that Bay, Row and Height will all be spaces.

11. ASCIIHEX PRESENTATION

The following integer values shall be sent and transformed to asciihex format before transferred over the interface.

ZZZZ	CRC (16-bit integer)	4 bytes
XX	Message Length (8-bit integer)	2 bytes
Counter	(8-bit integer)	2 bytes

Each byte is divided into 4 bit sections. Each section receives a corresponding ascii value.

Section hex	Ascii/hex value	Section hex	Ascii/hex value
0	0 0x30	8	8 0x38
1	1 0x31	9	9 0x39
2	2 0x32	A	A 0x41
3	3 0x33	B	B 0x42
4	4 0x34	C	C 0x43
5	5 0x35	D	D 0x44
6	6 0x36	E	E 0x45
7	7 0x37	F	F 0x46

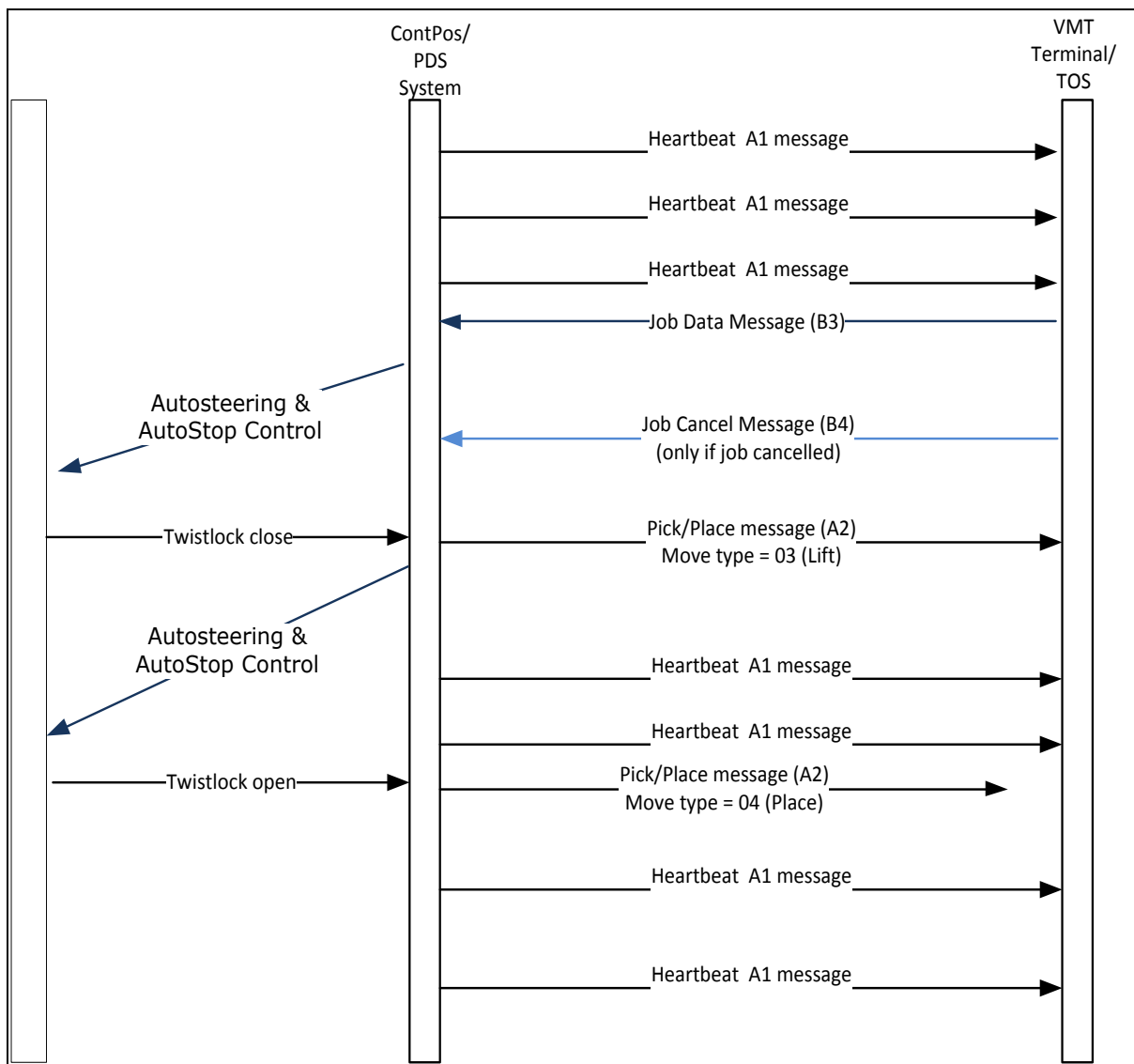
E.g. Counter value 100

The hex value is 0x64 and the asciihex value in the Counter field is 64 (0x36 0x34).

The heartbeat message size is 53 bytes. The value is 35 in hexadecimal format and the value in the message is presented in two bytes as (0x33 0x35).
The 16-bit integer is in big endian format, and the asciihex is in the same format, MSB section first.

12. MESSAGE SEQUENCES EXAMPLES

Message Interfaces as of example in RTG AutoStop/ContPos



Konecranes AutoStop message sequence example

13. VERSION HISTORY

Revision	Date	Description of change	Editor
1.00	5.9.2007	First version	MJM
1.01	9.7.2007	Integer fields converted to asciihex to support narrow band connections. Msgld's changed to ascii. Che id added to heartbeat message. System status bytes added at the end of heartbeat and pick/place message.	MJM
2.0	22.4.2009	Twist lock control functionality additions. New messages Job Twist lock Action request, Job Cancel Message and Job Continue Message. ACK message changed to include all messages, which requires acknowledgement. New file in message Heartbeat request and Heartbeat message	MJM
2.0b	13.12.2013	Word template changed to newest Konecranes template. Feature and organization names updated.	TG
2.1	17.04.2014	Ethernet support included. Language checked	MJM
3.0	25.04.2014	Heartbeat message field added. CHE position coordinates (PosX, PosY) + spreader and twist lock info	MJM
3.1	3.12.2014	Container position on truck added to job message B3	MJM
3.2	15.4.2015	Container weight added to A2 messages. Support for Gyro system removed.(Top-pick/Side-pick)	MJM
3.3	21.4.2015	Container position on truck added to A2 message	MJM

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